Student number:

Assignment 10 Semester 1, 2023

School of Mathematics and Statistics

MAST10012 Introduction to Mathematics

Submission deadline: 5:00 PM (17:00, AEST), Thursday 25 May 2023

This assignment consists of 11 pages (including this page)

Instructions to Students

Writing

- If you have a printer (or can access one), print the assignment one-sided. Write your answers in the boxes provided on the assignment that you have printed.
- If you do not have a printer, and you know how to annotate a PDF using a tablet, then annotate your answers directly onto the assignment PDF and save a copy for submission.
- If you cannot use the template, write your answers on blank A4 paper. Mimic the template by numbering your pages, and note which answers to write on each page.
- Record your student ID number in the box at the top of this cover page (or the first page of your submission). Do not record your name and tutorial details on this paper.
- Provide complete and fully justified solutions for each question.
- Write legibly, and use appropriate notation. Marks will not be awarded to unclear work.
- If you run out of space for a question, append additional work to the end of your submission. Make a note ("See page X") in the relevant question box warning the marker.
- Teaching staff cannot help you directly with assignment questions. Please ask on the Ed Discussion board if you need clarification on the wording of a question.

Scanning

- Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Do not scan on an angle. Crop images to the page borders. Make sure that your PDF file is readable. Material that cannot be read will not be marked.

Submitting

- Go to the Gradescope window. Choose the relevant Canvas assignment. Make sure that you upload the correct PDF file. You should get a Gradescope confirmation email.
- Assignments submitted late will incur a penalty of 4% (of the total mark available) for every hour or part thereof up to 24 hours. Written assignments cannot be submitted more than one day (24 hours) late.

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Question 1 (9 marks)

It may be assumed that dates of birth in a large population are distributed such that the probability of a randomly chosen person's date of birth being in December is 0.09. Let X be the number of people with a birthday in December.

(a) Find the probability that out of seven people chosen at random exactly two will have their birthday in December. Clearly show the distribution you use and show your working. Round your answer to four decimal places.

(b) Find the probability that out of nine people chosen at random at least one will have their birthday in December. Clearly show the distribution you use. Round your answer to four decimal places.

(c) In a group of m people, the probability that at least one person has a birthday in December is at least 0.7. What is the minimum number of people, m, we would need to randomly select so that the probability at least one person has a birthday in December exceeds 0.7? *Hint: the function* $y = \log_a(x)$ *is decreasing for* $0 < a < 1$, *so if* $x_1 < x_2$ *then* $\log_a(x_1) > \log_a(x_2)$.

Question 2 (13 marks)

The queueing time, X minutes, of a traveller at the ticket office of Melbourne Central train station has the probability density function f_X, defined by:

$$f_X(x) = \begin{cases} k(25x - x^3), & 0 \leq x \leq 5 \\ 0, & \text{elsewhere} \end{cases}$$

(a) Calculate the value of k.

(b) What is the probability that someone has to queue between 0 and 1 minutes, given that they queued for less than 2 minutes?

(c) Calculate the expected queueing time at Melbourne Central station (in minutes).

(d) Find the standard deviation of the queueing time at Melbourne Central station (in minutes). Round your final answer to two decimal places (use fractions or maintain more decimal places for previous steps).

Additional writing space for any question commences on the next page

Additional answer space for any question—submit this page even if blank

End of Assignment

Formula Sheet

Combinatorics

- $n! = n \cdot (n-1) \cdot (n-1) \cdots 3 \cdot 2 \cdot 1$
- $\binom{n}{k} = \dfrac{n!}{(n-k)! \cdot k!}$

Functions

- Lines: $y = mx + c$ Point-slope formula: $y = m(x-a) + b$
- Quadratics: $y = ax^2 + bx + c = \underbrace{a(x-r)(x-s)}_{\text{Factorised}} = \underbrace{a(x-h)^2 + k}_{\text{Turning point form}}$, use: $\frac{b}{2a}$
- Quadratic formula: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, check: $\Delta = b^2 - 4ac$

Exponentials

- $a^0 = 1$
- $a^1 = a$
- $a^{-1} = \dfrac{1}{a}$
- $a^{\frac{1}{2}} = \sqrt{a}$
- $a^x \cdot a^y = a^{x+y}$
- $\dfrac{a^x}{a^y} = a^{x-y}$
- $(a^x)^y = a^{x \cdot y}$
- $\dfrac{a^x}{b^x} = \left(\dfrac{a}{b}\right)^x$
- $a^x \cdot b^x = (a \cdot b)^x$
- $a^{\frac{x}{y}} = \sqrt[y]{a^x}$
- Change of base: $a^x = c^{x \log_c(a)}$ for any $c \in \mathbb{R}$

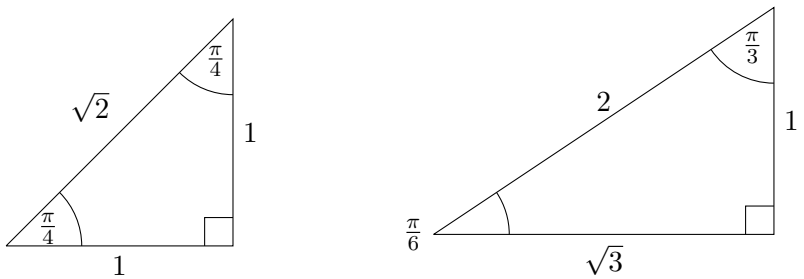
Logarithms

- $\log_e(x) = \log(x) = \ln(x)$
- $\log_a(1) = 0$
- $\log_a(a) = 1$
- $\log_a(p^x) = x \cdot \log_a(p)$
- $\log_a(p \cdot q) = \log_a(p) + \log_a(q)$
- $\log_a\left(\dfrac{p}{q}\right) = \log_a(p) - \log_a(q)$
- Change of base: $\log_a(p) = \dfrac{\log_c(p)}{\log_c(a)}$ for any $c \in \mathbb{R}$

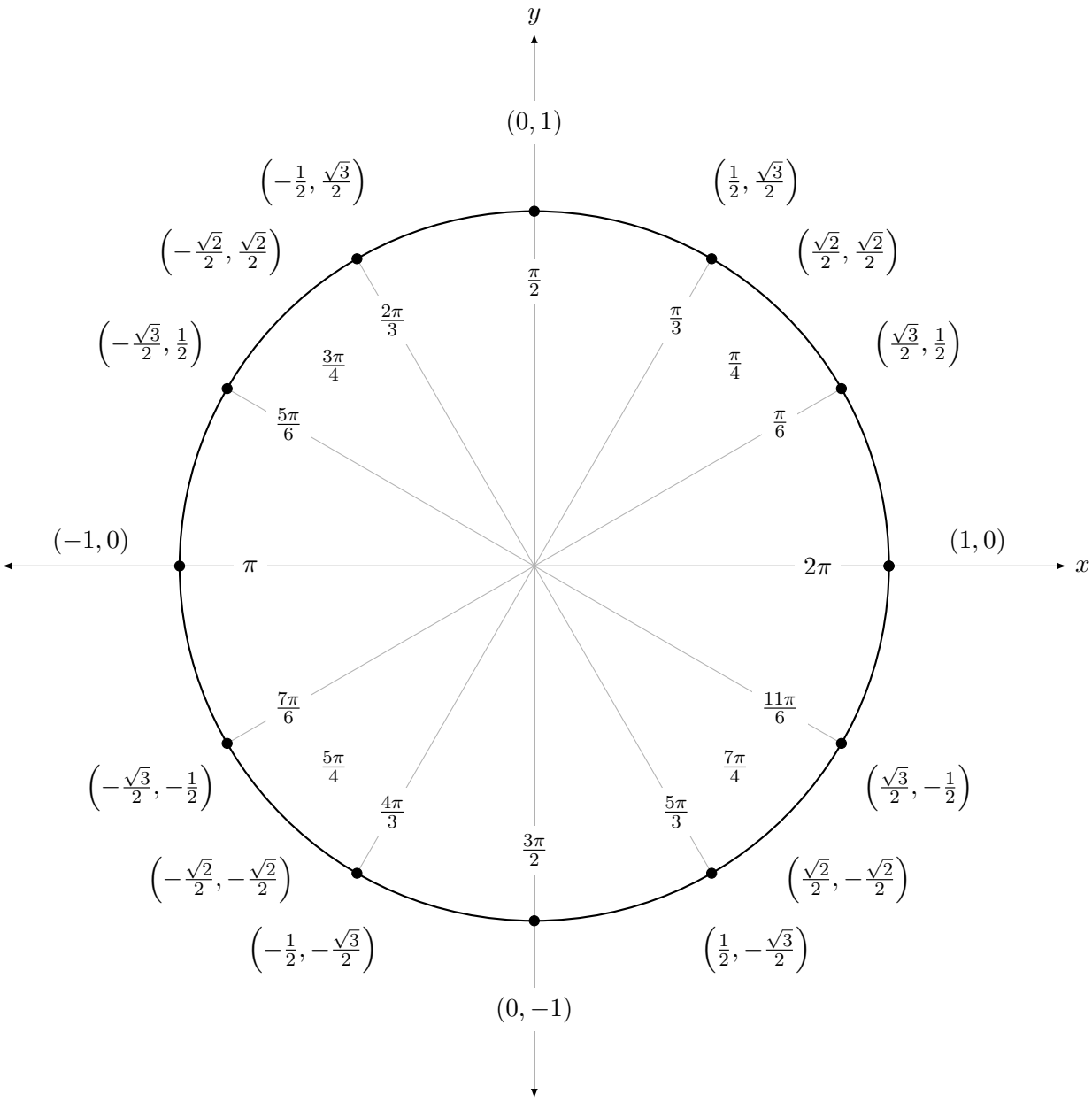
Trigonometric Identities

- $\sin(x + 2\pi) = \sin(x)$
- $\cos(x + 2\pi) = \cos(x)$
- $\sin(\pi - x) = \sin(x)$
- $\cos(\pi - x) = -\cos(x)$
- $\sin(-x) = -\sin(x)$
- $\cos(-x) = \cos(x)$
- $\sin(x + y) = \sin(x)\cos(y) + \cos(x)\sin(y)$
- $\cos(x + y) = \cos(x)\cos(y) - \sin(x)\sin(y)$
- $\sin(2x) = 2\sin(x)\cos(x)$
- $\cos(2x) = \cos^2(x) - \sin^2(x)$
- $\sin^2(x) = \frac{1-\cos(2x)}{2}$
- $\cos^2(x) = \frac{1+\cos(2x)}{2}$

Reference Triangles



Unit Circle



Limits

If we have $\lim_{x\to a} f(x) = L$, $\lim_{x\to a} g(x) = M$ and $c \in \mathbb{R}$ is a constant, then:

- $\lim_{x\to a} (f(x) + g(x)) = L + M$
- $\lim_{x\to a} (f(x) \cdot g(x)) = L \cdot M$
- $\lim_{x\to a} (f(x))^n = L^n$
- $\lim_{x\to a} c \cdot f(x) = c \cdot L$
- $\lim_{x\to a} c = c$
- $\lim_{x\to a} x^n = a^n$
- $\lim_{x\to a} (f(x) - g(x)) = L - M$
- $\lim_{x\to a} \left(\frac{f(x)}{g(x)}\right) = \frac{L}{M}$
- $\lim_{x\to a} (f(x))^{\frac{1}{n}} = \sqrt[n]{L}$
- $\lim_{x\to a} x = a$
- $\lim_{x\to a} x^{1/n} = \sqrt[n]{a}$

Calculus

- $f'(x) = \lim_{x\to a} \frac{f(x) - f(a)}{x - a} = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$
- $\frac{d}{dx}(c) = 0$
- $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$
- $\frac{d}{dx}(\log(x)) = \frac{1}{x}$
- $\frac{d}{dx}(\cos(x)) = -\sin(x)$
- $\frac{d}{dx}(x^n) = n \cdot x^{n-1}$
- $\frac{d}{dx}(e^x) = e^x$
- $\frac{d}{dx}(\sin(x)) = \cos(x)$
- $\frac{d}{dx}(\tan(x)) = \frac{1}{\cos^2(x)}$
- Product rule: $(f \cdot g)'(x) = \frac{d}{dx}(f(x) \cdot g(x)) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$
- Quotient rule: $\left(\frac{f}{g}\right)'(x) = \frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{(g(x))^2}$
- Chain rule: $(f \circ u)'(x) = \frac{d}{dx}(f(u(x))) = f'(u(x)) \cdot u'(x)$

Geometry

- For a triangle with base-length b and height h, the area is $\frac{1}{2}bh$.
- For a circle with radius r, the perimeter is $2\pi r$ and the area is πr^2.
- For a cylinder with radius r and height h, the volume is $\pi r^2 h$.
- For a pyramid or cone with base area A and height h, the volume is $\frac{1}{3}Ah$.
- For a sphere with radius r, the volume is $\frac{4}{3}\pi r^3$.

Probability

- $Pr(X \leq x) = 1 - Pr(X > x)$
- $Pr(B|A) = \frac{Pr(A \cap B)}{Pr(A)}, \quad Pr(A) \neq 0$
- For a discrete random variable X:
  $Pr(X \leq x) = \sum_{k=1}^{x} Pr(X = k)$

Binomial Distribution

For $X \overset{d}{=} \text{Bin}(n, p)$:

- $Pr(X = k) = \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k}$
- $\mathbb{E}(X) = n \cdot p$
- $\text{Var}(X) = n \cdot p \cdot (1-p)$

The Standard Normal Distribution Table

To standardise the normally distributed random variable X with population mean μ_X and population standard deviation σ_X, use:

$$X \stackrel{d}{=} N\left(\mu_X, \sigma_X^2\right) \quad \Rightarrow \quad Z = \frac{X-\mu}{\sigma} \stackrel{d}{=} N\left(0, 1^2\right)$$

For the standard normal distribution, $Z \stackrel{d}{=} N\left(\mu_Z = 0, \sigma_Z^2 = 1^2\right)$, the probability $Pr(Z \leq z) = Pr(Z < z)$ for a specified z is given to four decimal places in the below table. Note that z is specified to three decimal places, indicated in the row and column headers.

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999